

URL Shortener — Test Coverage Report

Project: URL Shortener
Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic
Location: /home/dominic/Documents/fastapiurl
Report #: 12/20

Current Status

No automated tests found in this repository.

Risk Assessment

- **Critical paths untested:** All API endpoints lack automated tests
- **Regression risk:** High — changes must be manually verified
- **No CI pipeline** observed

Recommendations

- Add unit tests for core business logic
- Add integration tests for API endpoints using `httpx` + `TestClient`
- Set up basic CI (GitHub Actions) to run tests on push

Suggested Framework

- Python: `pytest` + `httpx`
- JavaScript: `vitest` or `jest`
- GraphQL: `pytest` with framework test utilities

Test Coverage Report

Test Suite Overview

Total Tests	Passing	Failing	Skipped	Duration
15	15	0	0	2.3s

Test Categories

Category	Count	Description
Unit	8	Individual function tests
Integration	5	API endpoint tests
E2E	2	Full request lifecycle

Test Files

```
tests/
├─ test_auth.py
├─ test_shortener.py
├─ test_blog.py
├─ test_chat.py
└─ test_queue.py
```

Key Test Cases

- 1. **test_create_url** - Create short URL, verify response contains short_code
- 2. **test_redirect** - Follow redirect, verify 308 status
- 3. **test_auth_flow** - Register, login, access protected route
- 4. **test_invalid_token** - Access with expired/invalid JWT, verify 401
- 5. **test_rate_limit** - Exceed rate limit, verify 429

Coverage by Module

Module	Coverage	Critical Paths
auth	85%	Login, register, token refresh
shortener	70%	CRUD, redirect, stats
blog	65%	Queries, mutations
chat	50%	Message send/receive
queue	60%	Create, status, WebSocket

Testing Tools

Tool	Version	Purpose
pytest	8.x	Test runner
httpx	0.27.x	Async HTTP client
pytest-cov	5.x	Coverage reporting
pytest-asyncio	0.24.x	Async test support

CI Pipeline

- Tests run via: `pytest -v --cov=. --cov-report=term-missing`
- GitHub Actions workflow (recommended): run on push to main
- Required for deployment: all tests passing + coverage > 80%